

ASHUTOSH SHANKAR

Lead Data Scientist & AI/ML Engineering Leader • 15 Years Experience

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and AI/ML Engineering Leader with 15 years of experience defining, leading, and evangelizing analytics solutions across retail e-commerce. Proven expertise in end-to-end model development and deployment lifecycle, delivering mission-critical, user-centric solutions using Agile/Scrum methodology. Passionate about thought leadership and innovation across NLP, GenAI, LLMs, Agentic AI, and full-stack AI product development.

Domain Focus: Search Optimization · AI Stylist & Recommendation Systems · Visual Vector Search · Email Marketing · Pricing & Promo Optimization · Customer Sentiment Analysis

TECHNICAL SKILLS

NLP & GenAI	LLM Interaction, Entity Extraction, Text Summarization, GloVe, FastText, BERT, Hugging Face, LangChain, LangGraph, Agentic AI, CrewAI, MCP
LLMs	Amazon Bedrock (Claude Sonnet 4 via cross-region inference), OpenAI GPT, Google Gemini, Meta Llama, Mistral, Ollama, Phi-2, Fine-Tuning (LoRA, QLoRA, PEFT Adapter)
Computer Vision	CLIP (OpenAI), SentenceTransformers, Multi-modal visual vector search, Garment detection & hotspot mapping, Image-to-image & text-to-image retrieval
Machine Learning	Regression, Logistic Regression, Naive Bayes, SVM, Decision Tree, Random Forest, Clustering, PCA, Light GBM, Propensity Modeling, Churn Prediction
Deep Learning	Neural Networks, Keras / TensorFlow, CNN, RNN, LSTM, GRU, MLflow, Airflow ETL Pipeline
Frontend	React 18, Vite, Framer Motion, React Router, lucide-react, REST API integration
Backend / APIs	FastAPI, Python 3.9+, boto3, Uvicorn, RESTful API design
Cloud & DevOps	Amazon Bedrock, Amazon S3, AWS CLI, Vertex AI (GCP), Docker, Git / GitHub
Databases	SQL, Hive, MySQL, MongoDB, Cassandra, Pinecone, FAISS, ChromaDB (Vector DBs)
Languages	Python (NumPy, scikit-learn, Pandas), JavaScript, Jupyter
GitHub Projects	Sty-Lyst Chatbot (Bedrock + CLIP + React) · RL Tic-Tac-Toe · RASA Restaurant Chatbot · Hand Gesture Recognition · Fake Job Prediction
Leadership	Stakeholder Management, Team Hiring & Mentoring, Performance Appraisals, Agile / Scrum, Project Delivery

PROFESSIONAL EXPERIENCE

Lead Engineer (AI/ML) – JCPenney

July 2020 – Present

- Architected and built Sty-Lyst — a full-stack AI personal stylist chatbot for JCPenney — powered by Amazon Bedrock (Claude Sonnet 4) with three intelligent modes: Comp-Lyst (AI outfit builder generating structured 4-piece outfit JSON rendered as product carousels), Sty-Seek (CLIP-based visual vector search across 1.6 million+ catalog images), and Ask Me Anything (product Q&A grounded in spec sheets and customer reviews).
- Designed a multi-modal full-stack architecture: React 18 / Vite frontend, FastAPI backend, ChromaDB vector database, Amazon S3 image storage (37 clothing categories), and Amazon Bedrock LLM inference with cross-region inference profiles.
- Implemented CLIP-powered visual inspiration search enabling customers to upload outfit photos, detect individual garments via interactive AI hotspots, and find visually similar items from 1.6 million+ catalog images using text-to-image retrieval.
- Enhanced search result relevance by 15% for a major e-commerce platform by implementing ML and NLP techniques including FastText embeddings.

- Led Transformer / LLM / GenAI-based search optimization initiatives, engaging business stakeholders to maximize ML-driven business value in digital engineering.
- Fine-tuned large language models (LLMs) using QLoRA, LoRA, and PEFT Adapter techniques for domain-specific retail use cases.
- Drove adoption of Agentic AI frameworks (CrewAI, LangGraph, MCP) to automate complex analytical workflows.
- Managed talent acquisition, coaching, and performance evaluation processes for a data science team.

Senior Engineer 2 – JCPenney

September 2016 – June 2020

- Designed, developed, and deployed machine learning models end-to-end — from data collection to insight generation — for customer complaints and card fraud detection.
- Built propensity models for marketing campaigns enabling one-to-one personalised communication for each customer.
- Developed Divisional Propensity Models using Light GBM and Logistic Regression to predict customer affinity by category, generating decile rankings to support promotional campaign selection across channels.
- Built Churn Prediction Models using Light GBM to score active customers and identify high-risk segments for targeted retention marketing strategies.

Senior Engineer – Blue Yonder

June 2014 – August 2016

- Managed end-to-end project delivery, mentored team members, and contributed best practices and value-added improvements.
- Developed automation testing framework from scratch for warehouse labour management covering both frontend and backend validation.
- Recreated customer journeys using REST Assured and HTTP Client to perform root cause analysis (RCA).

Senior Engineer – Capgemini

December 2010 – May 2014

- Orchestrated a team focused on software development, product releases, and engineering tasks.
- Hired and trained staff, delegated work assignments, and collaborated across teams to identify and resolve technical issues.

EDUCATION

<i>2019 – 2021</i>	MSc in AI & Machine Learning	Liverpool John Moores University, London
<i>2019 – 2020</i>	Post Graduate Diploma in Machine Learning	IIIT Bangalore
<i>2006 – 2010</i>	BE in Electronics & Communication Eng.	Dayananda Sagar College of Engineering, Bangalore
<i>2002 – 2003</i>	Class XII – CBSE Board	
<i>2000 – 2001</i>	Class X – CBSE Board	

CERTIFICATIONS

▶ Agentic AI

•	Designing Agentic AI Systems on AWS	<i>Amazon Web Services (AWS)</i>	2025
•	Microsoft Certified: Agentic AI Business Solutions Architect	<i>Microsoft</i>	2025
•	AI Agents in LangGraph	<i>DeepLearning.AI + LangChain</i>	2025
•	Multi AI Agent Systems with crewAI	<i>DeepLearning.AI</i>	2025
•	The Complete Agentic AI Engineering Course	<i>Udemy</i>	2025

▶ LangChain

•	LangChain for LLM Application Development	<i>DeepLearning.AI / Coursera</i>	2025
•	Functions, Tools and Agents with LangChain	<i>DeepLearning.AI</i>	2025
•	Introduction to LangGraph	<i>LangChain Academy (LangChain Inc.)</i>	2025
•	AWS Generative AI Learning Plan for Developers	<i>Amazon Web Services (AWS)</i>	2025

•	IBM Generative AI Engineering Professional Certificate	IBM / Coursera	2026
▶	Vibe Coding		
•	Vibe Coding with GitHub Copilot	Scrimba / GitHub (Microsoft)	2025
•	GitHub Copilot Fundamentals	Microsoft / GitHub	2025
•	Vibe Coding Essentials – Build Apps with AI Specialization	Scrimba / Coursera	2026
•	Vibe Coding Camp: GitHub Copilot, Cursor, Lovable, Windsurf	Udemy	2025
•	Complete Prompt Engineering for AI Bootcamp	Udemy	2025
▶	Earlier Certifications		
•	Google Cloud – Generative AI	Google Cloud	2023
•	DeepLearning.AI – Generative AI with LLMs	DeepLearning.AI	2023
•	GCP – Cloud Digital Leader	Google Cloud	2022
•	Complete Prompt Engineering & AI Bootcamp	Udemy	2023
•	Neural Networks and Deep Learning	Coursera	2019
•	Machine Learning – Stanford University	Coursera / Stanford	2019
•	Machine Learning A-Z	Udemy	2019
•	Reinforcement Learning in Python	Udemy	2018

INTERESTS & HOBBIES

 **Fitness Enthusiast** — Passionate about strength training and maintaining peak physical and mental health. Strong believer that discipline in fitness builds the same resilience and consistency needed in high-performance engineering roles.

 **Passionate Cricketer** — Avid cricket player with a deep appreciation for team strategy, adaptability, and performing under pressure — values that translate directly into leading data science teams and delivering mission-critical AI solutions.